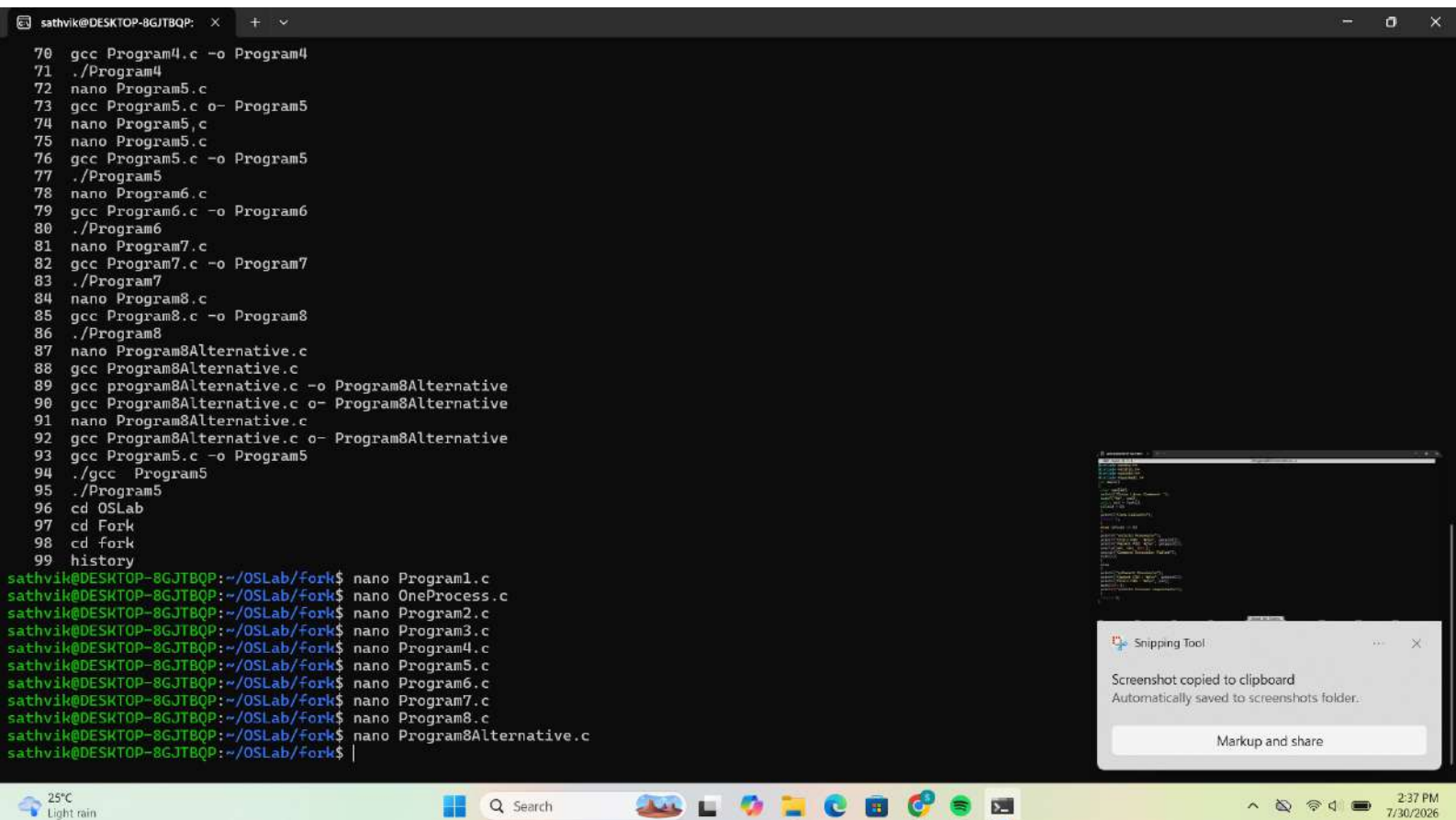




sathvik@DESKTOP-8GJTBQP: x + v

GNU nano 8.7.1Program8Alternative.c

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
int main()
{
    char cmd[50];
    printf("Enter Linux Command: ");
    scanf("%s", cmd);
    pid_t pid = fork();
    if(pid < 0)
    {
        printf("Fork Failed\n");
        return 1;
    }
    else if(pid == 0)
    {
        printf("\nChild Process\n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID: %d\n", getppid());
        execlp(cmd, cmd, NULL);
        perror("Command Execution Failed");
        exit(1);
    }
    else
    {
        printf("\nParent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);
        wait(NULL);
        printf("\nChild Process Completed\n");
    }
    return 0;
}
```

[Read 34 lines]

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute	^C Location	M-U Undo	M-A Set Mark	M-] To Bracket
^X Exit	^R Read File	^_ Replace	^U Paste	^J Justify	^_ Go To Line	M-E Redo	M-6 Copy	^B Where Was

25°C Light rain

Search

2:37 PM 7/30/2026

The screenshot shows a Windows desktop environment. At the top, a terminal window titled "GNU nano 8.7.1" displays a C program named "Program8.c". The code is as follows:

```
#include <stdio.h>
#include <unistd.h>
int main()
{
    int i;
    for(i=0;i<3;i++)
    fork();
    printf("PID = %d\n",getpid());
    return 0;
}
```

Below the terminal window, a taskbar is visible with several icons and a search bar. The system tray on the right shows the date and time as "2:36 PM 7/30/2026".

The screenshot shows a Windows desktop environment. At the top, there is a terminal window titled "GNU nano 8.7.1" with the file "Program7.c" open. The code in the terminal is as follows:

```

#include <stdio.h>
#include <unistd.h>
int main()
{
    pid_t pid = fork();
    if(pid == 0)
    {
        printf("Child executing ls...\n");
        execl("/bin/ls", "ls", "-l", NULL);
    }
    else
    {
        printf("Parent Waiting...\n");
    }
    return 0;
}

```

Below the terminal window, there is a taskbar with several icons: a weather icon showing "25°C Light rain", a search bar, and several application icons including a file explorer, a web browser, and a music player. On the right side of the taskbar, there is a system tray with icons for network, volume, and power, along with the date and time "2:36 PM 7/30/2026".

The screenshot shows a Windows 10 desktop environment. At the top, a terminal window titled "Program6.c" is open, displaying the following C code:

```
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
int main()
{
    fork();
    fork();
    printf("PID = %d\n", getpid());
    return 0;
}
```

Below the terminal window, a row of keyboard shortcuts is visible, including "Read 9 lines", "Execute Justify", "Location Go To Line", "Undo Redo", "Set Mark Copy", and "To Bracket Where Was".

The Windows taskbar at the bottom features the Start button, a search bar, and several pinned application icons. The system tray on the right shows the date and time as "2:36 PM 7/30/2026" and the weather as "25°C Light rain".

sathvik@DESKTOP-8GJTBQP: x + v

GNU nano 8.7.1Program5.c

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
int main()
{
    pid_t pid = fork();
    if(pid == 0)
    {
        printf("Child Executing\n");
    }
    else
    {
        wait(NULL);
        printf("Parent Executing\n");
    }
    return 0;
}
```

^G Help

^X Exit

^O Write Out

^R Read File

^F Where Is

^_ Replace

^K Cut

^U Paste

[Read 17 lines]

^T Execute

^J Justify

^C Location

^_ Go To Line

M-U Undo

M-E Redo

M-A Set Mark

M-6 Copy

M-] To Bracket

^B Where Was

25°C

Light rain

Search

2:35 PM

7/30/2026

The image shows a Windows desktop environment. At the top, a terminal window titled "GNU nano 8.7.1" is open, displaying a C program named "Program4.c". The code in the terminal is as follows:

```
#include <stdio.h>
#include <unistd.h>
int main()
{
    pid_t pid;
    pid = fork();
    if(pid == 0)
    {
        printf("Child PID = %d\n", getpid());
        printf("Parent PID = %d\n", getppid());
    }
    else
    {
        printf("Parent PID = %d\n", getpid());
        printf("Child PID = %d\n", pid);
    }
    return 0;
}
```

The terminal window has a title bar with standard Windows window controls (minimize, maximize, close) and a tab labeled "sathvik@DESKTOP-8GJTBQP: X". Below the terminal window, the Windows taskbar is visible. It includes a Start button, a search bar, and several pinned application icons: a globe, a folder, a shopping cart, a mail icon, a calendar, a music note, a speech bubble, and a power button. On the right side of the taskbar, there is a system tray with icons for network, volume, and battery, along with the date and time "2:35 PM 7/30/2026". Above the taskbar, a weather widget shows "25°C Light rain". A context menu is open over the taskbar, displaying various keyboard shortcuts and actions such as "Help", "Exit", "Write Out", "Read File", "Where Is", "Replace", "Cut", "Paste", "Execute", "Justify", "Location", "Go To Line", "Undo", "Redo", "Set Mark", "Copy", "To Bracket", and "Where Was".

The screenshot shows a Windows desktop environment. At the top, a terminal window titled "Program3.c" is open, displaying the following C code:

```
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
int main()
{
    pid_t pid;
    pid = fork();
    if(pid == 0)
        printf("I am Child\n");
    else
        printf("I am Parent\n");
    return 0;
}
```

Below the terminal window, a taskbar is visible with several icons: a weather icon showing 25°C and light rain, a search icon, and several application icons including a file explorer, a web browser, and a music player. To the right of the taskbar, a status bar displays the time as 2:35 PM and the date as 7/30/2026.

The screenshot shows a Windows 10 desktop environment. At the top, a terminal window titled "GNU nano 8.7.1" is open, displaying a C program named "Program2.c". The code in the terminal is as follows:

```
#include <stdio.h>
#include <unistd.h>
int main()
{
    fork();
    printf("Hello\n");
    return 0;
}
```

Below the terminal window, a taskbar is visible with several icons. On the left, there is a weather widget showing "25°C" and "Light rain". In the center, there is a search bar with the text "Search". To the right of the search bar, there are several application icons, including a file explorer, a web browser, and a music player. On the far right of the taskbar, there are system icons for network, volume, and battery, along with the date and time "2:35 PM 7/30/2026".


```
sathvik@DESKTOP-8GJT8QP: x + v
60 nano OneProcess.c
61 gcc OneProcess.c -o OneProcess
62 ./OneProcess
63 gcc Program2.c -o Program2
64 ./Program2
65 nano Program3
66 nano Program3.c
67 gcc Program3.c -o Program3
68 ./Program3
69 nano Program4.c
70 gcc Program4.c -o Program4
71 ./Program4
72 nano Program5.c
73 gcc Program5.c -o Program5
74 nano Program5.c
75 nano Program5.c
76 gcc Program5.c -o Program5
77 ./Program5
78 nano Program6.c
79 gcc Program6.c -o Program6
80 ./Program6
81 nano Program7.c
82 gcc Program7.c -o Program7
83 ./Program7
84 nano Program8.c
85 gcc Program8.c -o Program8
86 ./Program8
87 nano Program8Alternative.c
88 gcc Program8Alternative.c
89 gcc Program8Alternative.c -o Program8Alternative
90 gcc Program8Alternative.c -o Program8Alternative
91 nano Program8Alternative.c
92 gcc Program8Alternative.c -o Program8Alternative
93 gcc Program5.c -o Program5
94 ./gcc Program5
95 ./Program5
96 cd OSLab
97 cd Fork
98 cd fork
99 history
sathvik@DESKTOP-8GJT8QP:~/OSLab/fork$ |
```

```
sathvik@DESKTOP-8GJT8QP: x + v
30 #include <unistd.h>
31 int main()
32 { printf("PID = %d\n", getpid());
33 }
34 Only one process exists.
35 Introduce getpid().
36 uname -a
37 lscpu
38 lsblk
39 sudo apt install
40 lsblk
41 lsblk
42 lsblk
43 uname -a
44 lscpu
45 lsblk
46 ps -ef
47 top
48 free -h
49 uname -a
50 lsblk
51 cd OSLab
52 cd c
53 mkdir OSLab
54 cd OSLab
55 mkdir fork
56 cd fork
57 nano OneProcess
58 nano OneProcess.c
59 gcc OneProcess.c -o OneProcess
60 nano OneProcess.c
61 gcc OneProcess.c -o OneProcess
62 ./OneProcess
63 gcc Program2.c -o Program2
64 ./Program2
65 nano Program3
66 nano Program3.c
67 gcc Program3.c -o Program3
68 ./Program3
69 nano Program4.c
70 gcc Program4.c -o Program4
```

```
sathvik@DESKTOP-8GJTBQP: x + v
sathvik@DESKTOP-8GJTBQP:~$
sathvik@DESKTOP-8GJTBQP:~$
sathvik@DESKTOP-8GJTBQP:~$ cd OSLab
sathvik@DESKTOP-8GJTBQP:~/OSLab$ cd Fork
-bash: cd: Fork: No such file or directory
sathvik@DESKTOP-8GJTBQP:~/OSLab$ cd fork
sathvik@DESKTOP-8GJTBQP:~/OSLab/fork$ history
 1 sudo apt update
 2 sudo apt install build-essential
 3 gcc --version
 4 mkdir OSLab
 5 mkdir OSLab cd OSLab
 6 mkdir OSLab
 7 cd OSLab
 8 gedit hello.c
 9 pwd
10 cat process.c
11 pwd
12 touch hello.c
13 whoami
14 date
15 uptime
16 strace ls
17 sudo apt update
18 sudo apt install calendar
19 sudo apt update
20 sudo apt install ncal
21 which gcc
22 history
23 strace ls
24 sudo apt install strace
25 strace ls
26 history
27 calendar
28 nano fork.c
29 #include <stdio.h>
30 #include <unistd.h>
31 int main()
32 { printf("PID = %d\n", getpid());
```